

Cryptographic Chain-of-Custody for Last-Mile Parcel Delivery: A Blockchain–IoT System Architecture with Discrete-Event Simulation Evaluation

Author's Response

RESPONSE TO REVIEWERS

Manuscript ID: BLC-2026-04-0069.R1

Title: Cryptographic Chain-of-Custody for Last-Mile Parcel Delivery: A Blockchain–IoT System Architecture with Discrete-Event Simulation Evaluation

Author: Naushik Beladiya

Journal: IET Blockchain

We thank both reviewers for their careful reading of the manuscript. We are particularly grateful to Reviewer 1 for the detailed annotations in the marked-up manuscript, which identified a substantive error in our latency modelling and a serious error in the reference list. Both have been corrected. Correcting the latency model changed our results, and we have revised the abstract and conclusions accordingly rather than preserving the original claims.

Reviewer comments appear in italics. Changes are highlighted in the marked-up manuscript.

EDITOR-IN-CHIEF

If possible, please include relevant references from our journal that align with your research, to showcase an integration of existing knowledge from IET Blockchain.

We have added five references to work published in IET Blockchain, integrated where they bear on the argument rather than appended for completeness:

- Lin et al. (2024) on dual-blockchain logistics supply chain supervision [53], in Section 2.1. Their ledger-partitioning approach to the verifiability/confidentiality tension is contrasted with the payload-encryption approach taken here, and the trade-offs of each are stated.
- Haque et al. (2021), SmartOil [54], in Section 2.3, as a smart-contract supply chain system whose custody transitions are triggered by enterprise action rather than by an authenticated sensing device — the distinction on which this architecture turns.
- Augustine et al. (2026) on smart-contract-mediated IoT data sharing [55], in Section 2.3, whose taxonomy supports our claim that existing schemes address access control over collected data rather than attesting the collection event.
- Dixit et al. (2022) on cyber attacks against blockchain-based IoT systems [56], in Section 3.2. Their layered attack taxonomy corroborates our adversary classification, and their observation that device-layer compromise defeats rather than triggers cryptographic verification maps directly onto our class A3 and the limitation acknowledged in Section 7.1.
- Andrian and Wiputra (2026) on batching and off-chain storage [57], in Section 2.6, supporting the digest-anchoring decision in Section 6.4 while noting that their batching result does not transfer, since custody events must be anchored promptly to preserve timestamp evidentiary value.

REVIEWER 1 — SEMANTIC COMMENTS

S1. Latency calculation, TPS figures, and rollup type

On the latency calculations of the simulation (in public blockchains, TXs are coming from everywhere not just your system and hence latency time should be calculated according to the average waiting time for the TX in the minting mempool); also, the TPS numbers are not accurate. The kind of rollup commonly used in this problem (ZK vs. optimistic) is better mentioned and would be more accurate to use its specific metrics. This should be adjusted first before drawing any conclusions on latency.

We accept this criticism in full. The reviewer identified a genuine modelling error, and correcting it changed our results.

(a) The modelling error.

The original simulation computed confirmation time as the sum of a fixed mempool delay, drawn from $U(0.6, 2.2)$ seconds and applied identically to all three configurations, and a residual wait for the next block, drawn from $U(0, b)$. This formulation contains two defects. First, applying a single mempool distribution to every configuration is incorrect: contention is a property of the chain, and a permissioned deployment has no exogenous queue at all. Second, and more seriously, the $U(0, b)$ term assigns probability one to inclusion in the very next block, so no transaction ever competes for block space. This is precisely the closed-system assumption the reviewer identifies.

(b) The corrected model.

Block inclusion is now modelled as $U(0, b) + (G - 1) \cdot b$, where $G \sim \text{Geometric}(p)$ and p is the per-block probability of inclusion given the transaction's fee bid relative to competing demand under the EIP-1559 mechanism. The expected number of additional blocks waited is $(1 - p)/p$. Setting $p = 1$ recovers the closed-system case exactly, which is why the original formulation remains correct for Hyperledger Fabric and is retained there. For Ethereum mainnet, p is varied across three contention scenarios. The revised model is documented in Section 6.1, and the code is in the public repository.

(c) Reporting across contention scenarios.

Because p is not directly observable, we report results across three scenarios — benign ($p = 0.90$), moderate ($p = 0.50$), and congested ($p = 0.25$) — rather than at a single assumed value. We state explicitly in Section 6.1 that these are defined scenarios spanning a plausible operating range and not empirical measurements, and we identify empirical calibration of p against historical mempool data as future work in Section 8. We considered fitting p to observed mempool data for this revision, but concluded that presenting a transparent sensitivity range was preferable to a single empirically-derived point estimate whose provenance a reader could not easily audit.

(d) Revised results.

Table 4 has been replaced. The Ethereum median rises from 8.46 s (benign) to 14.01 s (moderate) to 31.12 s (congested), with the proportion of events confirmed within 15 seconds falling from 90.6% to 52.4% to 26.8%. Tail behaviour degrades considerably faster than the median: the 99th percentile reaches 193 s under congestion, against 15.9 s in the original model.

(e) Rollup family.

We agree this was an important omission, and addressing it improved the paper. We now specify that we model an optimistic rollup, the family in commercial logistics deployment, and Section 2.5 contrasts it with ZK rollups. This has a consequence material to the paper's central claim. Under an optimistic rollup, sequencer-level confirmation is soft finality resting on a trust assumption regarding a centralized operator, whereas hard finality — the point at which a custody record becomes censorship-resistant evidence — requires L1 settlement plus expiry of the seven-day fraud-proof window. For a system whose value is evidentiary, reporting only the sequencer figure would overstate the guarantee. Table 4 and Section 6.3 now distinguish the two.

(f) Corrected throughput figures.

Table 1 has been revised throughout, and now separates observed throughput from theoretical capacity:

- Ethereum mainnet: revised to 15–30 TPS observed, cited to the Ethereum developer documentation rather than to the whitepaper [51].
- Layer-2: the previous claim of approximately 40,000 TPS for Arbitrum described theoretical sequencer capacity, not realised throughput. Table 1 now reports 40–60 TPS observed against a theoretical ceiling of roughly 2,000–6,000 TPS.
- Hyperledger Fabric: revised from '3,500+ TPS' to the 500–3,000 TPS range documented in published Fabric 2.5 benchmarks, cited to the official performance documentation [52].
- A mean inclusion latency row has been added, as requested in the annotation on page 11.
- The 33% threshold is now qualified per deployment: 33% of total staked ETH on L1, 33% of the rollup's own stake on L2, and 33% of consortium membership in a permissioned deployment.

(g) Downstream corrections.

All conclusions dependent on latency have been re-derived: Table 4, Section 6.3, Section 8, and the abstract. The abstract's previous claim of sub-15-second confirmation for 94% of events has been replaced with the corrected range. We note that this is a material weakening of the headline result, and we consider it the correct outcome: the 94% figure was an artefact of the closed-system assumption. The updated abstract will be entered on the ScholarOne cover page and all associated forms.

S2. 'Optimization' should be described as a design decision with security impact

I think what is called optimization should be described as a design decision that has an impact on security (if offline data were altered).

Agreed, and we thank the reviewer for the correction. The original framing implied a strict improvement where a trade-off exists.

The two variants have been renamed throughout — in Section 4.3, Section 6.4, Tables 5 and 6, and Appendix A — from 'naïve' and 'edge-optimized' to 'on-chain-verified' and 'edge-attested'. Section 6.4 now includes an explicit analysis of the security difference. Moving digest computation and business-rule pre-validation to the device relocates part of the verification trust boundary to the edge: under the edge-attested variant, an adversary who has compromised a device (class A3) can submit a self-consistent digest over fabricated telemetry together with a compliance flag that the contract accepts without independent evaluation. We state this cost alongside the 25.1% gas saving, identify PUF-based device attestation as the mitigation, and specify that rules carrying regulatory consequence should remain on-chain regardless of gas cost. The choice between variants is now presented as a deployment-specific risk decision rather than a uniform recommendation.

S3. Threat model comments in the manuscript

The threat model have some comments in the manuscript.

Section 3 has been substantially rewritten. Specifically:

- Prose. We accept that the section read as summary bullets rather than journal prose. It has been rewritten as full exposition.
- Worked transaction. As requested in the annotation on page 13, the section now includes a symbolic custody transaction with each field defined, and maps each adversary class onto that structure explicitly.
- Dolev–Yao grounding. The model is now attributed to Dolev and Yao [46], with the two extensions required for the cyber-physical setting stated explicitly.
- Passive versus active attacks. These are now separated, with the rationale that passive attacks against a public ledger violate no protocol property and are therefore structurally undetectable, making their likelihood and appropriate defenses quite different from active attacks. A new adversary class A5 (Passive Observer) has been added, using the competitor-inference scenario the reviewer describes.
- MEV and Flashbots. We accept that the original framing was not well motivated. It has been rebuilt as an explicit contrast within class A5: the exposure here is competitor traffic analysis rather than value extraction, it is narrower in scope than MEV because randomized proposer selection prevents reliable positioning and the gain accrues only to an adversary with domain knowledge, and it is treated as a residual limitation of public deployment rather than a solved problem.
- Censorship. Adversary class A4 has been rewritten to foreground censorship over reordering, on the grounds that a validator or sequencer withholding custody events cannot create false records but can prevent timely creation of true ones — which for an evidentiary system negates the guarantee itself. Severity is graded across the three deployment options.
- 33% threshold. Now qualified per deployment, as under S1(f).
- Section order. The problem statement now precedes the threat model, per the annotation on page 14; the subsections have been renumbered accordingly.

S4. Relation to prior limitations and the TradeLens outcome

I advise referring to the limitations section in [Transportation Engineering 2025] and [Systems 2026, 14(5) 467, why Maersk failed] to discuss how part of their limitations and/or future work is handled in this paper.

We thank the reviewer for these pointers, which allow us to state our contribution against a concrete baseline rather than in the abstract.

Mutambik (2026).

Section 2.2 now engages directly with this paper's stated limitations [53]. It identifies last-mile vulnerability as under-addressed in the cold chain literature and acknowledges it as a limitation, and further notes that its blockchain and IoT components are treated as enabling architectural layers such that its results validate analytical feasibility rather than end-to-end operational validation of the digital components. The present work addresses both: last-mile custody is our explicit scope, and the blockchain–IoT layer is the object of quantitative evaluation rather than an assumed substrate.

Mofatteh and Fatahi Valilai (2025).

Discussed under W2 below.

TradeLens.

Section 2.5 now discusses the consortium governance failure. TradeLens processed over 30 million shipping events yet was discontinued in 2022, in significant part because competing carriers proved unwilling to commit operational data to a platform governed by a direct competitor. We make explicit why the proposed architecture does not inherit this failure mode: under public or L2 deployment no single participant owns the ledger, so participation does not require trusting a competitor's governance. We consider this a stronger justification for the platform choice than the one previously offered. We have cited the substance of this point rather than the vendor blog referenced in the review, which we did not consider an appropriate source for this journal.

S5. Censorship of traffic data; IoT devices producing wrong data with valid signatures

Then also discuss the possibilities and effect of: censoring traffic data, IoT producing wrong data with valid signature (their malfunctioning/fault tolerance); how the system will react, whether these are limitations.

Both are now addressed explicitly. We are grateful for the pressure on the second point in particular, since it bounds what our security claim can mean.

Sensor faults.

A correctly authenticated device reporting a false reading cannot be detected cryptographically: a signature attests provenance, not truth. The system will accept and anchor such a reading, and no property of the architecture prevents it. This is now stated in three places rather than left implicit. Section 3.2 records it as an explicit scope assumption. A new subsection in Section 7.1 treats it as a stated boundary of the security claim, and observes that for an evidentiary system, immutably recording a false reading is arguably worse than not recording it, since the record carries an unwarranted appearance of authority. Section 8 identifies cross-sensor plausibility checks, redundant multi-device attestation, and in-contract anomaly detection as future work; as the reviewer observes, the smart contract layer is the natural site for such logic.

Censorship.

Addressed within adversary class A4 as described under S3, with severity graded across L1, L2, and permissioned deployments and connected to the censorship-resistance row of Table 1.

REVIEWER 1 — WRITING AND REFERENCE COMMENTS

Additional correction: the centralized-baseline detection rate (pp. 21–22)

Explain from where comes this ratio. ... again not clear where this ratio came from. What is written before did not clarify why the TDR can't [be forged] in the blockchain case.

The reviewer is right that the baseline figure was asserted without a reproducible basis. We have addressed this by implementing the centralized baseline as an explicit detector over the same injected attack stream, rather than stating a figure in prose. Under the manuscript's own threat model — an administrator-privileged insider (A2) who can alter both custody records and the audit log — the only tampering that cannot be silently concealed is a

signature-forgery event carrying a device key absent from the registry, which trips a commit-time foreign-key constraint. Computed across the 30 seeds, this gives a baseline TDR of $19.9\% \pm 0.2\%$ (95% CI), equal within sampling error to the 20% signature-forgery share of the attack mix. The previous figure of 12.4% is withdrawn; the corrected value and its derivation now appear in Section 6.2.1, and the deficit relative to the blockchain system is restated as 80.0 percentage points in Section 6.2. The baseline detector is included in the public repository (baseline_model.py) so the figure is reproducible.

On the reviewer's related question — why the corresponding attack cannot succeed in the blockchain case — Section 6.2 now states the mechanism explicitly: a forged event fails ECDSA verification before block inclusion, and once a legitimate event is recorded in a finalized block its signed fields cannot be altered, so detection does not depend on the good behaviour of any administrator.

W1. Missing citations for added material

In many added parts, like case study examples, AES-GCM, Dolev Yao model and other parts details are just written without citing any reference for each case.

Accepted without reservation. Primary references have been added throughout:

- Dolev–Yao adversary model: Dolev and Yao (1983), IEEE Trans. Inf. Theory [46]
- AES-GCM: NIST SP 800-38D [47]
- SHA-256: FIPS PUB 180-4 [48]
- ECDSA: FIPS PUB 186-5 [49]
- Key management: NIST SP 800-57 Part 1 Rev. 5 [50], cited in Sections 5.1 and 7.3
- Ethereum platform mechanics: Ethereum developer documentation [51], replacing the bare citation to the whitepaper
- Hyperledger Fabric: official Fabric performance documentation [52]

Case study material in Section 2.5 is now supported by the industry survey reference [44] where no direct primary source is available, per the reviewer's guidance on page 11.

The ratios the reviewer queried on pages 19, 21 and 22 concern the centralized-baseline detection rate; these are addressed in the dedicated subsection above, where the figure is now computed reproducibly rather than asserted.

W2. Reference [33] and a full audit of the reference list

Disturbing issue of fake or maybe mixed-up original reference [33] ... your description fit ref.[33] of the 2026 survey I suggested in the previous report (M. Mofatteh O. Valilai, 'A blockchain ecosystem model for a sustainable last-mile delivery operation management', 9/2025, ScienceDirect - Transport Engineering). Please Correct this.

The reviewer is correct, and we apologise. The citation as printed does not correspond to a real publication. The work described is Mofatteh, M. Y., and Fatahi Valilai, O. (2025), 'A blockchain ecosystem model for a sustainable last-mile delivery operation management', Transportation Engineering. This has been corrected in Section 2.4 and in the reference list.

Rather than correct this entry in isolation, we treated it as grounds to audit the entire reference list against published records. That audit identified further problems, which we report in full:

- [27] attributed the PackChain paper to the wrong authors and gave an inexact title. Corrected to El Moudaa, Ibrahim, Kadadha, Mizouni, Otrouk and Singh, 'PackChain: Toward a Blockchain-Based Management Platform for Last-Mile Delivery'. The discussion in Section 2.3 has been revised accordingly.
- [44] (previous numbering) carried an incorrect title. Corrected to Cerchione, Passaro and Tavano, 'The Integration of AI, Blockchain and IoT for the Sustainable Development of the Logistics Service Industry: Insights From a PRISMA-Based Analysis', Sustainable Development (2026).
- Four further entries — the previous [20], [28], [49] and [51] — could not be verified against any published record. Rather than retain citations we could not confirm, we have removed all four and rewritten the sentences that depended on them (Sections 2.1, 2.2, 2.3).

- Three entries — the previous [40], [41] and [42] — were drawn from a separate research project of the author's on freight brokerage technology adoption and had no bearing on the present work. These have been removed along with the sentence in Section 7.4 that cited them.

The reference list now contains 52 entries. Every in-text citation resolves to a listed reference, every listed reference is cited at least once, and the numbering is contiguous. We regret these errors and thank the reviewer for the scrutiny that surfaced them.

W3. Abstract

The abstract is better formed and clarified in this version (but must correct the latency part after delay recalculations). After that, the abstract needs to be updated on the cover (and all other places where it is written in the journal forms).

The abstract's latency claims have been revised to match the corrected model, as described under S1(g). The updated abstract will be entered on the ScholarOne cover page and in all associated journal forms at submission.

REVIEWER 1 — STRUCTURAL AND EDITORIAL ANNOTATIONS

The remaining annotations in the marked-up manuscript have been addressed as follows:

- Page 6: the reviewer correctly notes that [1] and [2] are the Bitcoin and Ethereum whitepapers and do not support a claim about last-mile loss exposure. The preface now cites [3] and [4], the sources that do; [1] and [2] are introduced in Section 2.1, where they belong.
- Page 8: Section 2.1 now states what immutability contributes to traceability specifically — an append-only ledger replicated across mutually distrusting parties yields an activity log that is tamper-resistant and independently verifiable, so provenance claims need not be taken on the record-holder's word.
- Page 8: Section 2.1 now states what Zhu et al. (2022) improves over Tian (2016), namely smart-contract-enforced access control and on-chain certification state.
- Page 9: Section 2.2 now explains why device-level authentication is a precondition for the architecture rather than an adjunct to it.
- Page 9: Mutambik (2026) added as the 2026 follow-up, per S4.
- Page 9: the centralized-logging limitation is now stated explicitly in Section 2.2 as reintroducing at the storage layer the vulnerability the ledger removes.
- Page 11: Table 1 amended as described under S1(f).
- Page 11: EIP-1559 is now referenced in the simulation section, where the fee and block-size dynamics govern the inclusion model.
- Page 12: Table 2's performance-evaluation row has been adjusted, since the corrected [33] does employ numerical analysis and simulation. The table now differentiates on integrated IoT tamper detection, reported variance, and contention-scenario coverage rather than on the presence of simulation.
- The addition of adversary class A5 has been propagated: the abstract, Section 1, and Section 8 now refer to a five-class threat model.
- Page 24: addressed under S2.
- Page 28: conclusions re-derived from the corrected results, per S1(g).

REVIEWER 2

I thank the authors for the thorough and well-organized revision. The point-by-point response is clear, and the changes are easy to trace in the highlighted manuscript. All of the concerns I raised in my first review have been fully addressed. The manuscript is substantially improved, and I have no further comments.

We thank Reviewer 2 for the positive assessment and for the first-round comments that shaped the previous revision.